

 सत्यमेव जयते	भारत सरकार दूरसंचार विभाग दूरसंचार अभियांत्रिकी केंद्र Government of India Department of Telecommunications Telecommunication Engineering Centre K.L. Bhawan, Janpath, New Delhi- 110001	 ISO 9001:2015
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No. 26-1/2022-FA/TEC

Date: 04.09.2025

Subject: Minutes of 5th meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group (SG)-15 held on 04.09.2025.

Fifth (5th) meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group 15 regarding “*Networks, Technologies, and Infrastructures for Transport, Access & home*”, reconstituted for the Study Period 2025-28, was held on 04.09.2025 under the chairmanship of Sh. Rakesh Desai, DDG(FA), TEC through video-conferencing. The list of participants is placed at Annexure A.

2. After formal welcoming of the participants, Deo Pratap, AD (FA) briefed the participants about the agenda and further proceedings of the meeting.

3. Sh. Rakesh Desai, Chair/ NWG-15 & DDG (FA) TEC, welcomed all participants and briefly explained the members the schedule of the second meeting of ITU-T SG15 being held at ITU-T HQ, Geneva from 13-24 October, 2025 and submission deadlines. He also apprised the members about examination of the finalised contributions of NWG-15 by 3-member internal committee and the contributors have to modify their contributions as per the suggestions of this committee, if any, and the competent authority at DOT HQ will decide to accord their approval considering the recommendations of the committee. He also emphasized the need to attend online correspondence discussion meetings and interim Rapporteur’s group meeting (RGM). The Chair also underlined the requirements of preparing contributions considering the scope of the respective question and highlighting detailed gap analysis of the proposed work-item, bringing out the status of global standards in the relevant technology.

After the introduction, Sh. Rakesh Desai explained the members on potential work-items for contributions and also requested members to actively participate in the discussions on various contributions.

4. **Discussions were held on various potential work-items and contributions presented by the NWG-15 members as under: -**

Sl. No.	Contribution/ Potential Work Item Title and Question Number	Contributors / Experts	Suggestions from Members / Way forward
1	Proposal for a new work item L.fifomcf on “optical Fan-In Fan-Out device (FIFO) for multi-core optical fibre (MCF) and cable” (Q7/15)	Experts from Academia from IIT-Madras viz. Prof. Deepa, Balaji Srinivasan, Mr. Sobhanbabu; Expert from SFO- Mr Samuel Varghese & Mr Manjunath, Sh.Arvind Mishra from STL Sh.Rakesh Desai from TEC.	- Content of the documents is well improved since last meeting due to frequent discussions in the focus group meetings. - IEC SC86/WG7 meeting documents may be examined and suitably incorporated in the contribution, if any. - A.1 justification may be improved. - Draft to be improved considering other suggestions of the members.
2	Proposal to modify the L.400/L.12 or new recommendation / discussions for multicore fibre splices. (Q7/15)	Prof. Deepa, Mr. Sobhanbabu and Dr. Chandan Singh Yadav from IIT-Madras.	- Content may be prepared considering the scope of the existing recommendation and examining the TR.SDM - IEC SC86B document on optical splicing may be examined. - Draft to be improved considering other suggestions of the members.
3	Indian experience of the deployed Multicore optical fibre cable and test results. (Q5/15)	Experts from Academia from IIT-Madras, Sh.Arvind and Sh. Sudipta Bhaumik from STL.	- Confidential / business sensitive information may be removed in consultation with M/s STL. - Results may be explained in Tabular form also. - Draft to be improved considering other suggestions of the Members.

Sl. No.	Contribution/ Potential Work Item Title and Question Number	Contributors / Experts	Suggestions from Members / Way forward
4	Test Results and measurement methods for cross talk parameters of single-mode weakly-coupled multi-core optical fibre and cable. (Q5/15)	Prof. Deepa, Mr. Sobhanbabu and Dr. Chandan Singh Yadav from IIT-Madras, Sh.Arvind and Sh. Sudipta Bhaumik from STL, Sh.Rakesh Desai.	- As agreed by Indian experts in Q5/15 meeting on G.smmcf, Inputs on test results may be shared jointly by DOT, IIT-M and STL. - Last date of the correspondence activity ends at 12 September, 2025.
5	Draft Revision of Recommendation G.665 “Generic characteristics of Raman amplifiers and Raman amplified subsystems”- Adding new the new Appendix IV describing Raman amplifiers safety mechanisms (Q6/15)	Experts from Academia viz. IIT-M, IIT-Hyderabad.	- Draft contribution may be prepared considering the latest working document of the Editor on this work-item, prepared after June 2025 Paris meeting. - As this work-item is targeted for consent at the next SG15 Plenary meeting in October 2025, status of corresponding discussion for this work-item may be ascertained.
6	Addition of an annexure titled “Broad level integration of MTN with multi-technologies” in G.8310 (2020) Amd 2. (Q12/15)	Ms.Divya Sharma and Sh. Jainendra Bahadur from NSCS, Sh. Deepika Ambwani and Sh. Ashish Kumar Pandey from CDOT, Delhi	- Contribution may be drafted based on the informative / non-normative modifications. - Proper justification and gap analysis may be brought out in the proposal part. - Draft to be improved considering other suggestions of the members.
7	Proposal for new draft ITU-T recommendation	Ms.Divya Sharma and Sh. Jainendra Bahadur from	- Proper justification and gap analysis may be brought out in the proposal part. Texts

Sl. No.	Contribution/ Potential Work Item Title and Question Number	Contributors / Experts	Suggestions from Members / Way forward
	L.arch_WDT “Architecture of Wavelength detector and tuning device to be used in Multi PON network”. (Q7/15)	NSCS, Sh. Ashish Kumar Pandey from CDOT, Delhi	like ‘Cost effective’ to be removed and to be replaced by proper term. - Draft to be improved considering other suggestions of the members.
8	Draft New Recommendation G.fso “Terrestrial free space optics for mobile backhaul with short reach interfaces” (Q6/15)	Sh.Rakesh Desai & Ms. Mudita Chandra from TEC, Sh. Gyan Deep Verma, Mr Abhishek Dixit / Ms.Anjali Gupta / Ms. Anupama Sharma from IIT- Delhi.	- Focus group of NWG-15 has discussed and prepared the draft texts on fibre length (Clause 6.2.1) and the power outage probability definition. - Last date of the correspondence activity ends at 5 th September 2025. - This work-item is targeted for consent at the next SG15 Plenary meeting in October 2025.
9	L.nis-Practical considerations for network infrastructures sharing (Q7/15)	Sh.Rakesh Desai & Ms. Mudita Chandra from TEC and Sh. Rajesh Kumar Soni-DDG (CS) DOT HQ, Mr Abdul from Airtel, Experts from TSPs/IPs, other experts of NWG- 15.	- Focus group of NWG-15 has discussed and prepared the draft texts on sharing of passive infrastructure and power supply systems. - Last date of the correspondence activity ends at 19 September, 2025.
10	Draft New Recommendation G.dfos “Distributed fibre optic sensing system for terrestrial optical transmission system” (Q6/15)	Experts from Industries (JMV LPS Ltd, STL), Sh. Vineet Malik, Director (Radio), TEC, Sh.Rakesh Desai-TEC.	- M/s JMV has shown interests in submitting inputs on case studies for backscattering and forward type sensing. - Sh. Vineet Malik, Director (Radio), TEC, is

Sl. No.	Contribution/ Potential Work Item Title and Question Number	Contributors / Experts	Suggestions from Members / Way forward
			preparing inputs on Energy aspects of DFOS for submission in correspondence discussions considering the discussions in the focus group meeting held on 2nd September 2025. - Last date of the correspondence activity ends at 12th September 2025.
11	G Supplement LLSOT(W)DM: Low Layer Split Mobile Fronthaul over T(W)DM PON-Revision G989.3 (NG-PON2) as a Fronthaul Technology to Support Coordinated Multi-Point (CoMP) in 6G (Q2/15)	Prof. Goutam Das, Preeti Samhita Pati, Sourabh Bhandarkar and Sourav Dutta from IIT Kharagpur, Atri Mukhopadhyay from IIT Bhubaneswar, Sh.Rakesh Desai-TEC.	- Focus group of NWG-15 are updating the draft as per the guidance by the Editor of G Supplement LLSOT(W)DM for submitting additional information so that Indian proposals can be incorporated in the TD. - Q2/15 next RGM and a joint meeting of Q2,3 and 4/15 is scheduled on Sep 8-11 2025 and 16 September 2025 respectively. Deadline for contributions is Friday before the meeting.
12	Contributions during correspondence discussions of Q5/15 work-items viz. Revision of L.102, L.Suppl.39, New L.ocwp, L.Suppl.ne. (Q5/15)	Sh.Rakesh Desai & Ms. Mudita Chandra from TEC NWG-15 Members, Sh.Bikram Nayak and Sh. Sudipta Bhaumik from STL	- Ms. Mudita on behalf of focus group explained the progress in these work-items. - Draft contributions are being updated considering the respective working documents and considering the RGM report of Q5/15.

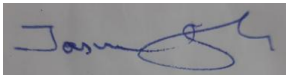
The Chair thanked all the contributors for presenting their contributions before NWG-15 members and stated that only existing contributions / work-items will be discussed in the next NWG-15 meeting in view of the prescribed timelines. Contributors and NWG-15 members were further requested to prepare afresh contribution(s) and / or improve the presented contribution(s) by 15th September 2025. Contributors and NWG-15 members were also requested to submit further inputs during correspondence discussions, adhering to the timelines prescribed by the Editor / Rapporteur of the respective work-item.

5. Sh. Rakesh Desai once again requested the NWG-15 members, not having TIES account, to obtain TIES account by registering on ITU portal and also requested all members to start access of ITU-T SG15 documents / recommendations, study it for gap analysis and prepare contributions to address such gaps. He also requested all the members of NWG-15 to register on TEC SCP (Standard Coordination Portal) and start accessing meeting documents from this portal.

6. The key action points emerged from the meeting are as under: -

- i. Members / experts are requested to submit contributions as per the timeline as discussed above.
- ii. Next (6th) meeting of NWG-15 will be held tentatively at 11 am on Wednesday, 17th September 2025. Meeting link will remain same i.e <https://cdotmeet.cdote.in/vmeet/rooms/1bb-a8c-qqt-ucn/join> .
- iii. Members, who are interested to contribute in any of the ongoing work-item and to become a members of particular focus group, are requested to convey their willingness to the Chair or the Convener, NWG-15.
- iv. Members are requested to register for the forthcoming ITU-T SG15 meeting, if not done.

7. The meeting ended with the vote of thanks to the chair.



(Jasvir Singh Panesar)
Convener, NWG-15 & Director (FA), TEC

To,

All the participants / Members of the NWG-15.

Copy to: 1. Advisor (T), DOT HQ, New Delhi. for kind information.

2. Sr.DDG & Head, TEC for kind information.

Annexure- 'A'**List of Participants**

1. Sh. Rakesh Desai, DDG (FA) & Chairman/ NWG-15, TEC
2. Sh. Jasvir Singh Panesar, Director (FA) & Convener, TEC
3. Sh. Vinod Kumar, DDG (Tx), TEC
4. Ms. Mudita Chandra, ADG(Tx), TEC
5. Sh. Avadhesh, RTEC
6. Sh. Arjun Singh, DoT
7. Sh. Rahul Chouhan, DOT
8. Sh. Ashish Kumar Pandey, CDOT Delhi
9. Sh. Sudipta Bhaumik (STL), Sterlite Technologies Ltd
10. Sh. Rajeev Chaturvedi, Dura-line India Pvt Ltd
11. Sh. Abdul Ashfaq, Bharti Airtel Limited
12. Sh. Jonaq Niveer Sarma, Indian Institute of Technology Madras
13. Dr. Partha Sarathi Das, Haldia Institute of Technology
14. Sh. Sameer, IIT Madras
15. Sh. Jainendra Bahadur, NSCS
16. Sh. Sudipto Das, Phoenix Contact India
17. Sh. Aneesh Sobhanan, IIT Hyderabad
18. Sh. Samuel Varghese, SFO TECHNOLOGIES PVT. LTD
19. Ms. Deepa Venkitesh, IIT Madras
20. Sh. Sobhanbabu, IITM
21. Anupma, IIT Delhi
22. Sh. Rana Pratap, IIT Madras, Academia
23. Sh. Chandan Singh Yadav, IIT Madras
24. Sh. Jayveer, Delhi Test House
25. Sh. K.S.P. Singh, PHOENIX CONTACT (INDIA) PVT. LTD
26. Sh. Bikram Kesharee Nayak, Sterlite Technologies Limited
27. Dr. Lenin Raja, AA Electro Magnetic Test Laboratory Priv
28. Sh. Gyan Deep Verma, IIT Delhi
29. Ms. Alka Kumari, Intern, TEC
30. Sh. Deo Pratap, AD (FA), TEC
